Fakulteit Ingenieurswese, Bou-omgewing & IT

Faculty of Engineering, Built Environment & IT

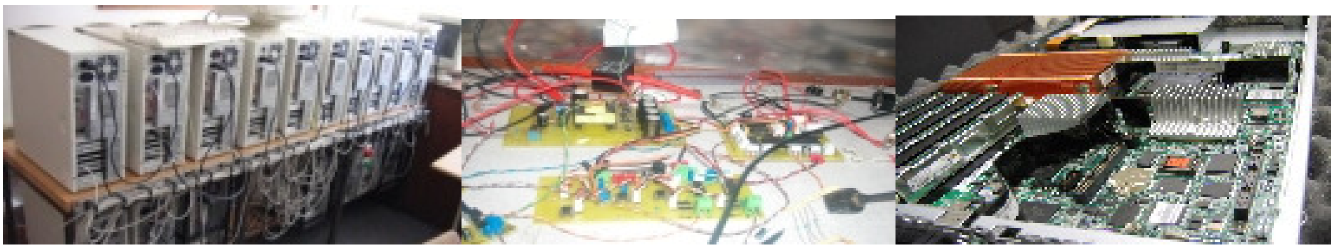
School of Engineering

Department of Electrical, Electronic and Computer Engineering

Electricity and Electronics EBN122

Lecturers: **Prof J Joubert and Mr D le Roux**

Last revision: 2013-06-20



UNIVERSITEIT VAN PRETORIA
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YUNIBESITHI YA PRETORIA
Denkleiers • Leading Minds • Dikgopolo tša Dihlalefi

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ORGANISATIONAL COMPONENT

1. GENERAL PREMISE AND EDUCATIONAL APPROACH

The general objective of this module is to develop expertise in solving electric and electronic circuits. The module is primarily focused on solving new problems based on known theoretical techniques, as opposed to the memorising of recipe-type procedures, or solving problems by simply inserting values into formulae. A thorough knowledge of the fundamental theory of circuits will be required, although the gaining of theoretical knowledge as such is not the primary objective of this module. Students will continuously be presented with problems that they might not have seen before, but for which they are adequately equipped to solve. Students are strongly advised to gain as much experience as possible in solving circuit problems from the beginning.

2. LECTURERS, VENUES AND CONSULTING HOURS

	Name	Room number and building	Telephone number and e-mail address
Lecturers	Prof J Joubert (Afr. Class)	1-4 Compact Antenna Range (Building 9), South Campus	(012) 420-2680 jjoubert@up.ac.za
	Mr D le Roux (English Class)	14-16 Eng 1	(012) 420-2201 derik.leroux@up.ac.za
Assistant Lecturers	Rogero dos Santos, Retha-Mari Green, Larry Schmidt	rogerioma7@gmail.com rethamari@gmail.com	
Secretary	Me C Freislich	13-20 Eng 1	(012) 420-3735 cornel.freislich@up.ac.za

Location of the laboratory
All practicals take place in the CAEC lab, room 4-39, Eng 2 building.

Notice board
All notices will be communicated by means of ClickUP.

Consulting hours
Consulting hours of lecturers, tutors and teaching assistants will be announced at the beginning of the semester, and will also be displayed on their office doors. Students may consult lecturers, tutors and teaching assistants only during the consulting hours as indicated, **or by appointment**. This policy also holds before tests and examinations. In other words, lecturers, tutors and teaching assistants are only available during their normal consulting hours on the days preceding tests and examination. This policy aims to encourage students to plan their work and to work continuously.

3. STUDY MATERIALS AND PURCHASES

Prescribed textbook: Charles K Alexander, Mathew N O Sadiku, *Fundamentals of Electric Circuits*, Fifth Edition, McGraw-Hill, 2013, ISBN: 9781259071393.

Equipment: Each student requires his/her own set of cables for the practicals. The cost of the cables is included in the class fees. (Arrangements will be made for students to receive their cables at the first practical session).

4. LEARNING ACTIVITIES

4.1 Contact time and learning hours

Number of lectures per week: Three
Number of tutor classes per week: One
Laboratory work: Three experiments of three hours each

This module carries a weighting of 16 credits, indicating that on average, a student should spend some 160 hours to master the required skills (including time for preparation for tests and examinations). This means that on average you should devote some 10 hours of study time per week to this module. The scheduled contact time is approximately five hours per week, which means that another five hours per week of own study time should be devoted to the module.

4.2 Lectures

Reading work that has to be completed prior to a particular lecture will appear on the Schedule (to be posted on the module website). The lecturer assumes that students have worked through the relevant material as presented in the textbook prior to the lecture. This is necessary as circuit explanations, which are often very mathematical in detail, can be difficult to follow without prior exposure to the material.

*If a student has a problem that was not solved during the lecture, he/she is welcome to make an **appointment** with the lecturer for consultation. Lectures will however not be repeated during consulting hours. Students must bring their own attempts at solving the problem to the **appointment**.*

4.3 Tutorial classes

The *Schedule* with problems to be worked out for each tutorial class will be posted on the module web page after the start of the second semester.

Students should solve the prescribed problems *before* the tutorial class. These problems will be discussed during the tutorial class; students should use the class to check if and where they erred in solving problems.

Apart from the formal tutorial sessions, tutors will also be available for **consultation by appointment.**

4.4 Laboratory (practical) work

Rules and requirements: Each student must do three practicals. In order to pass this module,
- all practicals must be done, and
- an average mark of 40% must be obtained for the practicals.

Laboratory groups and sessions: Students will be divided into large practicum groups at the start of the semester. The dates and times each group has to do a particular practical will appear on the course website. The responsibility rests on each student to make sure when practicals are scheduled for him/her. **Permission to be moved from a practicum group must be obtained in writing from the assistant lecturers at least one week before the date of the practical. Please see the web page for more information.**

Information regarding practicals: Specific information about practicals will be posted on the website prior to the commencement of the particular practical.

During laboratory sessions: Students must be well prepared when reporting for a laboratory session. The information on the website as well as the relevant material in the textbook must be consulted beforehand. *Each student must bring his/her own textbook to the practical.*

Absence from laboratory sessions: If a student was absent from a laboratory session, he/she must apply **within seven days of the date of the session** on an official form available from **assistant lecturers** to do a special laboratory session. An official and valid excuse (e.g., a medical certificate) will be required. The special laboratory session will be done at the end of the semester and the work will be the same as the session missed by the student.

Practical Assignments

1. **Practical 1:** Familiarization with simulation of DC circuits.
2. **Practical 2:** DC circuit analysis.
3. **Practical 3:** Construction and testing of OP AMP circuits

5. RULES OF ASSESSMENT

See also the examination regulations in the Year Books of the Faculty of Engineering, Built Environment and Information Technology (Part 1: Engineering, or Part 2: Built Environment and Information Technology).

Pass requirements: To pass this module, a student must
1. attain a final mark of at least 50%, and
2. do all practicals, and
3. attain a subminimum of 40% for the practicals.

Calculation of the final mark

Semester mark:	50%
Examination:	50% (The final examination lasts three hours)

Calculation of the semester mark:

Semester Test 1:	35%
Semester Test 2:	35%
Mini tests:	15%
Practicals:	15%

Examination admission: A semester mark of 40% is required for examination admission. Furthermore, all practicals must have been completed, and an average mark of 40% must have been obtained for the practicals.

Semester tests: Two tests of 90 minutes each will be written during the scheduled test weeks of the School of Engineering.

First test week:	17 – 24 Aug
Second test week:	5 – 12 Oct

Dates, times and venues will be posted on the Departmental website as soon as the test timetables become available.
Only two weeks will be afforded for queries and corrections on a test and/or the mark list once a semester test has been returned to students. After the two weeks, the mark that appears on the module web page for the relevant semester test will be considered as final.

Mini tests: Three mini (short) tests will be written in the course of the semester. The dates for the tests will be announced in class and on the website. *Only two weeks will be afforded for queries and corrections on a test and/or the mark list once a mini test has been returned to students. After the two weeks, the mark that appears on the module web page for the relevant mini test will be considered as final.*

Absence from tests and examinations:
The Departmental procedures concerning absence from formal evaluation opportunities (i.e. tests and examinations) as described in the EEC Guide will be followed in all cases of absence from examinations, tests and practical work. The EEC Guide is available at:
http://www.ee.up.ac.za/main/en/undergrad/guides.

Semester tests: If a student was absent from a semester test, he/she must apply **within seven days of the date of the test to write a special semester test**. **An official application form** for the special semester test **along with an official and valid written excuse** (e.g., a medical certificate) must be handed to **Ms Cornel Freislich (Room 13-20, Eng 1; 420-3735)**. The official application form is available at Ms Cornel Freislich's office. The special semester test for all rightful absentees will be written at the end of the semester and will deal with all the work covered in the module up to that point. If a student cannot complete the application form in person within seven days due to the circumstances that led to absence from the semester test, the Department should be informed by telephone, or a representative should be sent to fill in the application.

Mini tests: A student who is absent from a mini test loses the marks for the test. **An official and valid written excuse** (e.g., a medical certificate) must be handed to **Ms Cornel Freislich (Room 13-20, Eng 1; 420-3735)**. in case of a **valid absence**. The student's mini test mark will then be calculated from the marks for his/her other mini tests (i.e., there are no "supplementary" mini tests). If a student misses all mini tests with a valid excuse in each case, the contribution to the semester mark from the mini tests will be set (proportionally) equal to the contribution of the lowest of the two semester test marks.

6. GENERAL

Grievance procedures

The student may see the lecturer by appointment to make him/her aware of any problems. Any grievances that the student wishes to bring to the attention of the Department should be reported to the class representative who will pursue the matter further.

Plagiarism warning

Students are encouraged to discuss work with each other. However, each student should hand in his/her **own** work for assignments. Plagiarism, which also includes copying the work of another student during tests and exams and copying from the Internet, can lead to expulsion from the University.

Even if another student gives you permission to use his/her assignments or other research to hand in as you own, you are not allowed to do it. It is a form of plagiarism. You are also not allowed to let anybody copy your work with the intention of passing it off as his/her own work.

Speak to your lecturer if you are uncertain about what is required. For more information, see *http://www.ee.up.ac.za/main/en/undergrad/guides* or consult the brochure available at the Academic Information Service.

A statement regarding the originality of your work must be appended to ALL written work submitted for evaluation in this module. The statement can be found at *http://www.ee.up.ac.za/main/en/undergrad/guides* .

STUDY COMPONENT

1. MODULE OBJECTIVES, ARTICULATION AND LEARNING OUTCOMES

1.1 General objectives

The goal of this module is to equip the student with knowledge of and skill in the analysis of circuits containing combinations of resistors, operational amplifiers, capacitors and inductors, and the calculation of power. The student's understanding of the new concepts and the skill in their application will depend on how much practice the student gets in applying new concepts. It is essential that the student do as many as possible of the problems in each chapter in the textbook and that the student also consult other textbooks on the topic. The general objective with this module is to emphasise **understanding** rather than memorising. A problem-driven approach to learning is followed.

The module further accentuates the practical testing of theoretical principles and calculations through the measurements of real circuits. Students will gain experience in the use of instruments such as the multimeter and oscilloscope and will be given the opportunity to work with different types of power sources and active and passive circuit elements.

In order to achieve the objectives, attendance of and meaningful participation during lectures, laboratory sessions and tutor classes are essential. Furthermore, students are advised to embark on a well-structured and systematic study programme, in which the module material is studied in a probing, scientific and innovative manner, rather than by simple and passive memorising.

1.2 Articulation with other modules in the programme

This course is a base course for all three degrees of electrical engineering, electronic engineering and computer engineering, and a basic introduction to electric circuits for other engineering disciplines.

1.3 Critical learning outcomes

The following ECSA exit-level outcomes are addressed in the module, i.e. at the conclusion of this module the student will be capable of:

ECSA 2.1: **Problem solving**
Learning outcome: Demonstrate competence to identify, assess, formulate and solve convergent and divergent engineering problems creatively and innovatively.

ECSA 2.2: **Application of scientific and engineering knowledge**
Learning outcome: Demonstrate competence to apply knowledge of mathematics, basic science and engineering sciences from first principles to solve engineering problems.

ECSA 2.4: **Investigations, experiments and data analysis**
Learning outcome: Demonstrate competence to design and conduct investigations and experiments.

1.4 Cognitive level of assessment

		%*
1.	Knowledge	10
2.	Comprehension	10
3.	Application	60
4.	Analysis	10
5.	Synthesis	
6.	Evaluation	
7.	Other skills**	10

* Estimate of the % of the total assessment, including all forms of assessment applied in this module, devoted to the various levels of cognitive thinking skills and of "other skills".

** Assessment of "other skills":
Presentation skills
Report writing and language skills
Practical skills √
Team working skills √

See also paragraph 3.5 for an explanation of Bloom's classification of lower- to higher-order thinking skills.

2. MODULE STRUCTURE

Study theme and study units	*Mode of instruction*	*Chapter*	*Contact sessions*
1. Basic Concepts • Systems of Units • Charge and Current • Voltage • Power and Energy • Circuit Elements • Applications	Lectures, class discussion, tutor classes and self-study.	1	2
2. Basic Laws • Ohm's Law • Nodes, Branches, and Loops • Kirchhoff's Laws • Series Resistors and Voltage Division • Parallel Resistors and Current Division • Wye-Delta Transformations • Applications	Lectures, class discussion, tutor classes and self-study.	2	7
3. Methods of Analysis • Nodal Analysis • Nodal Analysis with Voltage Sources • Mesh Analysis • Mesh Analysis with Current Sources • Nodal and Mesh Analysis by Inspection • Nodal Versus Mesh Analysis • Transistors • Applications	Lectures, class discussion, tutor classes and self-study.	3	7

Study theme and study units	*Mode of instruction*	*Chapter*	*Contact sessions*
4. Circuit Theorems • Linearity Property • Superposition • Source Transformation • Thevenin's Theorem • Norton's Theorem • Derivations of Thevenin's and Norton's Theorem • Maximum Power Transfer • Applications	Lectures, class discussion, tutor classes and self-study.	4	6
5. Operational Amplifiers • Operational Amplifiers • Ideal Op Amp • Inverting Amplifier • Non-inverting Amplifier • Summing Amplifier • Difference Amplifier • Cascaded Op Amp Circuit • Applications	Lectures, class discussion, tutor classes and self-study.	5	5
6. Capacitors and Inductors • Capacitors • Series and Parallel Capacitors • Inductors • Series and Parallel Inductors • Applications	Lectures, class discussion, tutor classes and self-study.	6	5
7. Sinusoids and Phasors • Sinusoids • Phasors • Phasor Relationship for Circuit Elements • Impedance and Admittance • Kirchhoff's Laws in the Frequency Domain • Impedance Combinations • Applications	Lectures, class discussion, tutor classes and self-study.	9	8
8. Sinusoidal Steady-State Analysis • Nodal Analysis • Mesh Analysis • Superposition Theorem • Source Transformation • Thevenin and Norton Equivalent Circuits • Applications	Lectures, class discussion, tutor classes and self-study.	10	8
9. Laboratory work – 3 experiments	Computer-assisted and experimental work.		(3 x 3) = 9
		Total:	**57**

3. GUIDELINES FOR USING THE STUDY THEME DESCRIPTIONS

The information given in the next sections of this study manual under the various study theme headings is intended to assist students in acquiring the required skills and achieve the learning outcomes effectively. The following specific information is included under each of the study theme headings:

3.1 Learning outcomes of the study theme

The given learning outcomes for each study theme are essential to achieve the critical learning outcomes as set out in Section 1.3.

3.2 Study units

The title of the study unit and references to appropriate study material are given here. The study of the referenced study material is regarded as the minimum required to achieve the learning outcomes satisfactorily.

3.3 Self-study activities

Here information is given regarding exercises and problems related to the study material, which should be attempted and which are in accordance with the criteria of assessment of the study theme.

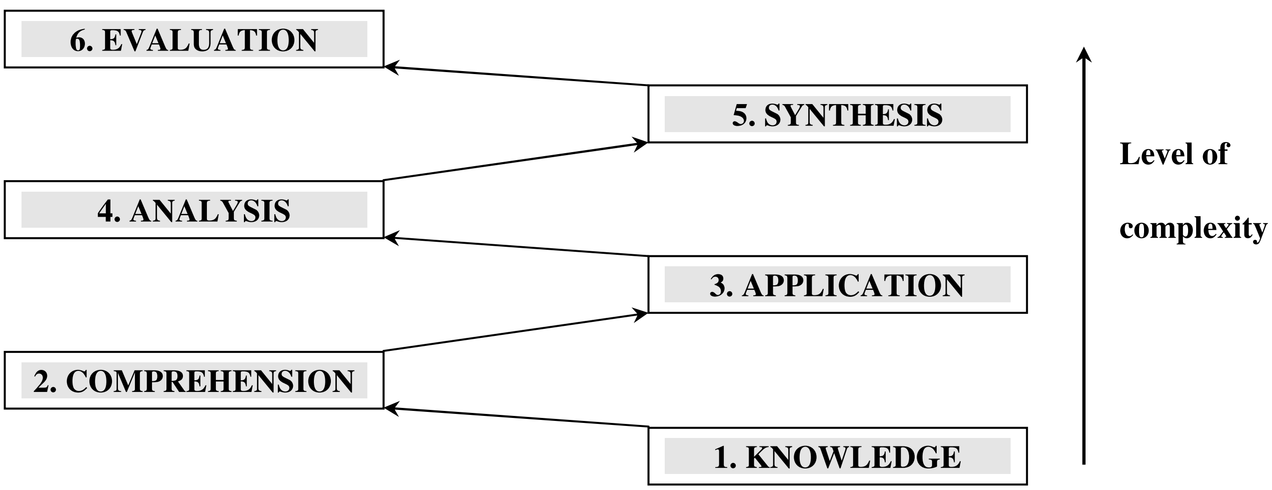
3.4 Assignments

Here information is given regarding assignments (tutorial problems).

3.5 Criteria of assessment

The criteria of assessment are a list of specific skills to be mastered by the student in order to achieve the learning outcomes of the syllabus theme. During assessment (tests and the examination), students will be evaluated in terms of these criteria.

The statements used to define the criteria of assessment are classified in terms of a series of lower- to higher-order thinking skills (cognitive domains), in accordance with Bloom's *Taxonomy of Educational Objectives* (Bloom BS and Krathwohl DR, *Taxonomy of educational objectives. Handbook 1. Cognitive domain,* Addison-Wesley, 1984):



The characterisation of the cognitive domains is given in the table below.

	Cognitive Domain	Definition	Typical Action Verbs
1.	Knowledge	Remembering previously learned information.	Arrange, define, describe, identify, label, list, match, name, outline
2.	Comprehension	Understanding the meaning of information.	Classify, discuss, estimate, explain, give example(s), identify, predict, report, review, select, summarise, interpret, "in your own words"
3.	Application	Using the information appropriately in different situations	Apply, calculate, demonstrate, illustrate, interpret, modify, predict, prepare, produce, solve, use, manipulate, put into practice
4.	Analysis	Breaking down the information into the component parts and seeing the relationships.	Analyse, appraise, calculate, compare, criticise, derive, differentiate, choose, distinguish, examine, subdivide, organise, deduce
5.	Synthesis	Putting the component parts together to form new products and ideas.	Assemble, compose, construct, create, design, determine, develop, devise, formulate, propose, synthesise, plan, discuss, support
6.	Evaluation	Making judgments of an idea, theory, opinion, etc., based on criteria.	Appraise, assess, compare, conclude, defend, determine, evaluate, judge, justify, optimise, predict, criticise

At the first-year level, a larger proportion of questions will be based on the lower levels (levels 1 to 3), whilst final-year examinations will contain a larger proportion of questions based on the higher-level thinking skills (levels 4 to 6).

4. STUDY THEME DESCRIPTIONS

4.1 STUDY THEME 1: Basic Concepts

4.1.1 Learning outcomes

The student will:

- Understand the basic SI units and SI prefixes
- Learn about the concepts of charge, current, and voltage
- Learn about power and energy
- Become familiar with the passive sign convention
- Learn about the circuit elements including dependent and independent sources

4.1.2 Study units

<u>Introduction</u>

Chapter 1, section 1.1

<u>SI units</u>
Chapter 1, section 1.2

<u>Charge and Current</u>
Chapter 1, section 1.3

<u>Voltage</u>
Chapter 1, section 1.4

<u>Power and Energy</u>
Chapter 1, section 1.5

<u>Circuit Elements</u>
Chapter 1, section 1.6

<u>Applications and Problem Solving (self study)</u>
Chapter 1, section 1.7 and section 1.8

4.1.3 Self-study activities

All examples, review questions and problems

4.1.4 Tutorial class problems

As listed in the *Schedule* on the website.

4.1.5 Criteria of assessment

Upon completion of this study theme, the student will:

- **know** the SI units and be able to **determine** derived units.
- know the different equations and units for voltage and current and be able to apply them.
- be able to **define** the basic circuit element.
- **know** the passive sign convention and be able to **apply** it.
- **know** the equation for power and be able to **apply** it.
- be able to **derive** the equation for power in terms of the basic circuit element's terminal voltage and current and be able to **apply** it.
- **know** the characteristics of ideal voltage and current sources.
- be able to **apply** and **interpret** the passive sign convention in power calculations.
- be able to **compute** charge, voltage, current, power and energy for an element.

4.2 STUDY THEME 2: Basic laws

4.2.1 Learning outcomes

The student will:

- Learn about Ohm's law
- Understand the concepts of nodes, branches, and loops
- Learn Kirchhoff's current and voltage laws
- Learn about combining resistors in series and in parallel

- Learn about the principles of voltage division and current division
- Become familiar *with wye-delta transformation*

4.2.2 Study units

Ohm's Law
Chapter 2, section 2.2

Nodes, Branches, and Loops
Chapter 2, section 2.3

Kirchhoff's Law
Chapter 2, section 2.4

Series Resistors and Voltage Division
Chapter 2, section 2.5

Parallel Resistors and Current Division
Chapter 2, section 2.6

Wyne-Delta Transformations
Chapter 2, section 2.7

Applications (self study)
Chapter 2, section 2.8

4.2.3 Self-study activities

All examples, review questions and problems

4.2.4 Tutorial class problems

As listed in the *Schedule* on the website.

4.2.5 Criteria of assessment

Upon completion of this study theme, the student will:

- **know** the five ideal circuit elements.
- be able to **apply** the definitions of ideal sources and hence **determine** the validity of connected sources.
- **understand** the way dependent sources are handled.
- **understand** the difference between passive and active elements.
- **know** Ohm's law and be able to **apply** it.
- **know** the equations for power in terms of resistance and conductivity and be able to **apply** them.
- be able to **construct** circuit models and **identify** shortcomings with respect to models of practical systems.
- be able to **identify** nodes, loops and meshes.
- be able to **define** direct current, series, parallel and other basic concepts.
- be able to **derive** equations for elements equivalent to elements connected in series and/or parallel.

- be able to **solve** problems involving voltage and current division.
- be able to **design** voltmeters and ammeters.
- **know** Kirchoff's voltage and current laws and be able to **apply** them.
- be able to **derive** equations for delta-to-Y conversions and **apply** them

4.3 STUDY THEME 3: Methods of Analysis

4.3.1 Learning outcomes

The student will:
- Learn about nodal and mesh analysis
- Understand the concept of supernode and supermesh
- Apply what is learnt to DC transistor circuits

4.3.2 Study units

Nodal Analysis
Chapter 3, section 3.2

Nodal Analysis with Voltage Sources
Chapter 3, section 3.3

Mesh Analysis
Chapter 3, section 3.4

Mesh Analysis with Current Sources
Chapter 3, section 3.5

Nodal and Mesh Analysis by Inspection
Chapter 3, section 3.6

Nodal Versus Mesh Analysis
Chapter 3, section 3.7

Applications: DC Transistor Circuits (self study)
Chapter 3, section 3.9

4.3.3 Self-study activities

All examples, review questions and problems

4.3.4 Tutorial class problems

As listed in the *Schedule* on the website.

4.3.5 Criteria of assessment

Upon completion of this study theme, the student will be able to:

- determine a set of independent simultaneous equations consisting of node-voltage equations and constraint equations from which a circuit can be solved.
- **identify** a supernode and **formulate** a new set of equations associated with the supernode.

- **determine a set of** independent simultaneous equations consisting of mesh-current equations and constraint equations from which a circuit can be solved.
- **identify** a supermesh and **formulate** a new set of equations associated with the supermesh.
- **identify** the constraint equations and **formulate** the new set of equations associated with dependant sources for the mesh-current method.

4.4 STUDY THEME 4: Circuit Theorems

4.4.1 Learning outcomes

The student will:

- Learn about theorems used for simplifying analysis
- Understand the principles of linearity and superposition
- Learn of about the concepts of source transformation and equivalence.
- Be able to use Thevenin's theorem to find a Thevenin equivalent circuit.
- Be able to use Norton's theorem to find a Norton equivalent circuit.
- Understand the concept of maximum power transfer to the load.

4.4.2 Study units

<u>Linearity Property</u>
Chapter 4, section 4.2

<u>Superposition</u>
Chapter 4, section 4.3

<u>Source Transformation</u>
Chapter 4, section 4.4

<u>Thevenin's Theorem</u>
Chapter 4, section 4.5

<u>Norton's Theorem</u>
Chapter 4, section 4.6

<u>Derivations of Thevenin's and Norton's Theorem</u>
Chapter 4, section 4.7

<u>Maximum Power Transfer</u>
Chapter 4, section 4.8

<u>Applications (self study)</u>
Chapter 4, section 4.10

4.4.3 Self-study activities

All examples, review questions and problems

4.4.4 Tutorial class problems

As listed in the *Schedule* on the website.

4.4.5 Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **use** the linearity property to **calculate** circuit variables
- **use** the superposition principle to **calculate** circuit variables
- **derive** a source transformation for a given circuit and **use** source transformation to **calculate** circuit variables.
- **calculate** Thévenin and Norton equivalent circuits by **determining** which of the following methods is applicable and then **apply** it:
    - **determine** V_{Th} and R_{Th} by **calculating** the open-circuit voltage and short-circuit current.
    - **apply** source transformations
    - **determine** R_{Th} by the method of deactivating independent sources
    - **calculate** R_{Th} by deactivating independent sources and adding a test source for circuits containing dependant sources only.
- **use** the superposition principle to **derive** Thevenin's and Norton's Theorems
- **derive** the equations associated with maximum power transfer and **apply** the underlying principles to other circuits.
- **apply** the method of superposition to linear circuits.

4.5 STUDY THEME 5: Operational amplifiers.

4.5.1 Learning outcomes

The student will:

- Understand what operational amplifiers are
- Learn about the ideal op amp
- Learn some op amp circuits including inverting, noninverting, summing, and difference amplifiers

4.5.2 Study units

<u>Operational Amplifier and Ideal Op Amp</u>
Chapter 5, section 5.2 and section 5.3

<u>Inverting Amplifier</u>
Chapter 5, section 5.4

<u>Non-Inverting Amplifier</u>
Chapter 5, section 5.5

<u>Summing Amplifier</u>
Chapter 5, section 5.6

<u>Difference Amplifier</u>
Chapter 5, section 5.7

<u>Cascaded Op Amp Circuit</u>
Chapter 5, section 5.8

<u>Applications (self study)</u>

Chapter 5, section 5.10

4.5.3 Self-study activities

All examples, review questions and problems

4.5.4 Tutorial class problems

As listed in the *Schedule* on the website.

4.5.5 Criteria of assessment

Upon completion of this study theme, the student will be able to
- **derive** transfer characteristics of basic op-amp circuits (e.g., inverting-amplifier circuit, summing-amplifier circuit, noninverting-amplifier circuit, and difference-amplifier circuit) from first principles.
- completely **solve** circuits in which no more than two op-amps are connected.

4.6 STUDY THEME 6: Capacitors and Inductors

4.6.1 Learning outcomes

The student will
- Learn about capacitors and capacitance
- Understand how to combine capacitors in series and parallel
- Learn about inductors and inductance
- Understand how to combine inductors in series and parallel

4.6.2 Study units

<u>Capacitors</u>
Chapter 6, section 6.2

<u>Series and Parallel Capacitors</u>
Chapter 6, section 6.3

<u>Inductors</u>
Chapter 6, section 6.4

<u>Series and Parallel Inductors</u>
Chapter 6, section 6.5

<u>Applications (self study)</u>
Chapter 5, section 6.6

4.6.3 Self-study activities

All examples, review questions and problems

4.6.4 Tutorial class problems

As listed in the *Schedule* on the website.

4.6.5 Criteria of assessment

Upon completion of this study theme, the student will be able to

- **solve** time-varying voltages and currents in circuits containing combinations of either inductors or capacitors.
- **Compute** energy and power associated with the inductors and capacitors in the above problems.

4.7 STUDY THEME 7: Sinusoids and Phasors

4.7.1 Learning outcomes

The student will:

- Understand the concepts of sinusoids and phasors
- Apply phasors to circuit elements
- Be introduced to the concepts of impedance and admittance
- Learn about impedance combinations

4.7.2 Study units

<u>Sinusoids</u>
Chapter 9, section 9.2

<u>Phasors</u>
Chapter 9, section 9.3

<u>Phasor Relationship to Circuit Elements</u>
Chapter 9, section 9.4

<u>Impedance and Admittance</u>
Chapter 9, section 9.5

<u>Kirchhoff's Laws in the Frequency Domain</u>
Chapter 9, section 9.6

<u>Impedance Combinations</u>
Chapter 9, section 9.7

<u>Applications (self study)</u>
Chapter 9, section 9.8

4.7.3 Self-study activities

All examples, review questions and problems

4.7.4 Tutorial class problems

As listed in the Schedule on the website.

4.7.5 Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **compute** the phasor representation of sinusoidal currents and voltages and also determine inverse phasor transforms.
- **compute** the rms-value of a periodic function.
- **Apply** Kirchhoff's Law in the frequency domain

STUDY THEME 8: Sinusoidal Steady-State Analysis

4.8.1 Learning outcomes

The student will:

- Apply previously learnt circuit techniques to sinusoidal steady-state analysis
- Learn how to apply nodal and mesh analysis in the frequency domain
- Learn how to apply superposition, Thevenin's and Norton's theorems in the frequency domain

4.8.2 Study units

<u>Nodal Analysis</u>
Chapter 10, section 10.2

<u>Mesh Analysis</u>
Chapter 10, section 10.3

<u>Superposition Theorem</u>
Chapter 10, section 10.4

<u>Source Transformation</u>
Chapter 10, section 10.5

<u>Thevenin and Norton Equivalent Circuits</u>
Chapter 10, section 10.6

<u>Applications (self study)</u>
Chapter 10, section 1

4.8.3 Self-study activities

All examples, review questions and problems

4.8.4 Tutorial class problems

As listed in the Schedule on the website.

4.8.5 Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **compute** the frequency domain equivalent circuit of any circuit containing sinusoidal sources and combinations of resistors, inductors, and capacitors.
- **solve** frequency domain circuits by using standard circuit techniques.